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Efficient Bayesian Inference for Stress-Strength Reliability in Gompertz Distribution via Ranked Set Sampling

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Abstract:

The Gompertz distribution, with its exponentially increasing hazard rate, is widely used in reliability engineering and survival analysis. However, estimating stress-strength reliability under this distribution has received limited attention. Conventional simple random sampling (SRS) is inefficient when precise measurements are costly or time-consuming. Ranked set sampling (RSS) addresses this limitation by using inexpensive ranking before measurement, extracting more information without increasing sample size. This study develops a Bayesian estimation of stress-strength reliability $R = P(X < Y)$ for the Gompertz distribution under RSS. Bayesian estimators are derived under a squared error loss function using the Metropolis-Hastings algorithm. The significance of this work is providing an efficient, low-cost, and high-precision method for reliability estimation when accurate measurement is expensive or difficult.

Keywords: Ranked Set Sampling, Gompertz Distribution, Stress-Strength Reliability, and Metropolis-Hastings Algorithm.

Mathematics Subject Classification (2020): 62Exx, 62F10, 62F15.

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1 Introduction

[Gompertz \(1825\)](#) discovered that after adulthood, the risk of death increases exponentially with age and applied this law to help insurance companies that needed reliable mortality tables but had limited, outdated data. Instead of relying solely on raw data, he proposed fitting a smooth exponential curve to mortality figures. This approach filled data gaps and enabled reasonable predictions of death rates for very old ages. The Gompertz distribution models time-to-event data with an exponentially increasing hazard rate and has been generalized to improve flexibility. In medicine, the Generalized Exponentiated Unit Gompertz Distribution by [Sindhu et al. \(2024\)](#) successfully modeled arthritic pain relief times. Additionally, the Exponentiated Generalized Exponential Gompertz Distribution by [Karimi Ezmareh and Yari \(2025\)](#) further advances reliability and survival analysis. The practical use of these models relies on robust statistical inference for complex data structures. This need is addressed by [Lv et al. \(2024\)](#), who developed inference methods for general progressive Type II censored competing risks samples.

Definition 1.1. A continuous random variable X on the interval $[0, \infty)$ has a Gompertz distribution with parameters $\beta > 0$ and $\gamma > 0$, denoted by $G(\beta, \gamma)$, if its probability density function (PDF) and cumulative distribution function (CDF) are respectively:

$$f(x) = \beta e^{\gamma x} e^{-\frac{\beta}{\gamma}(e^{\gamma x} - 1)}, \quad (1.1)$$

$$F(x) = 1 - e^{-\frac{\beta}{\gamma}(e^{\gamma x} - 1)}. \quad (1.2)$$

The stress-strength reliability $R = P(\text{Strength} > \text{Stress})$ is a critical metric for assessing safety margins and failure risks in engineering design. [Hakamipour \(2025\)](#) enhanced multicomponent system analysis by modeling dependent component strengths using copula functions. [Hudaverdi and Susam \(2023\)](#) addressed scenarios with two correlated stress sources via a bivariate copula-based framework. [Karimi Ezmareh and Yari \(2022\)](#) derived inference procedures for stress-strength reliability under the Gompertz distribution with Type II censoring. These advances enable more realistic reliability assessment under complex dependencies.

Ranked Set Sampling (RSS), introduced by [Li et al. \(1952\)](#), is an efficient technique when measurement is costly but ranking is cheap. It involves visually ranking small random subsets and then precisely measuring only the top-ranked unit from each subset. This structured approach minimizes sampling variability and provides more informative data. As a result, RSS yields significantly more precise parameter estimates than Simple Random Sampling (SRS) with the same measurement effort.

To construct an RSS sample of size n , draw m independent sets of size m via SRS, rank units within each set without precise measurement, and select the r -th ranked unit from the r -th set. This yields one cycle of RSS with observations $X_{(1)1}, \dots, X_{(m)m}$. Repeating the procedure across k independent cycles gives a total sample of size $n = mk$, denoted as $X_{(1)1c}, \dots, X_{(m)mc}$ for $c = 1, \dots, k$. For convenience, we write X_{rc} in place of $X_{(r)rc}$.

$$f_{r:m}(x) = \frac{m!}{(r-1)!(m-r)!} [F(x)]^{r-1} [1 - F(x)]^{m-r} f(x), \quad (1.3)$$

$$F_{r:m}(x) = \sum_{i=r}^m \binom{m}{i} [F(x)]^i [1 - F(x)]^{m-i}. \quad (1.4)$$

Estimating stress-strength reliability using RSS represents a significant advancement in reliability engineering, as RSS's structured data collection mechanism systematically captures greater information from the population compared to SRS. [Basikhasteh et al. \(2020\)](#) successfully addressed the complexities of estimating R for a bathtub-shaped lifetime distribution by developing a Bayesian framework within a maximum RSS design. [Aldrabseh et al. \(2025\)](#) established that even for the classical exponential distribution, sophisticated RSS variants yield substantial improvement in estimation accuracy.

Accurate estimation of stress-strength reliability in Gompertz models remains challenging due to the statistical limitations of conventional sampling methods. Traditional reliability assessments using SRS often fail to achieve statistical efficiency, particularly when dealing with exponentially-increasing hazard rates and resource-intensive measurements. Our research addresses this limitation by integrating RSS within a Bayesian paradigm.

2 Bayesian Estimation of Parameters

In this section, Bayesian estimation of the Gompertz distribution parameters is obtained based on SRS and RSS. A Gamma prior distribution is used, and estimates are obtained under the squared error (SEL) loss function.

2.1 Estimation Based on SRS

Assume β has a Gamma prior with parameters (a_1, b_1) , and γ has a Gamma prior with parameters (a_2, b_2) . The joint posterior density is proportional to the product of the likelihood and prior:

$$\pi(\beta, \gamma \mid X) \propto \beta^{n+a_1-1} e^{-b_1\beta} \gamma^{a_2-1} e^{-b_2\gamma} e^{\gamma \sum x_i} e^{-\frac{\beta}{\gamma} \sum (e^{\gamma x_i} - 1)}. \quad (2.1)$$

The conditional posterior densities are:

$$(\beta \mid \mathbf{x}, \gamma) \sim \text{Gamma} \left(n + a_1, b_1 + \frac{1}{\gamma} \sum_{i=1}^n (e^{\gamma x_i} - 1) \right), \quad (2.2)$$

$$\pi(\gamma \mid \mathbf{x}) \propto \gamma^{a_2-1} e^{-\gamma(b_2 - \sum x_i)} e^{-\frac{\beta}{\gamma} \sum (e^{\gamma x_i} - 1)}. \quad (2.3)$$

Although samples for β are straightforward to draw, generating samples for γ requires the Metropolis-Hastings algorithm.

2.2 Estimation Based on RSS

Suppose we have an RSS of size $n = mk$, denoted by X_{ic} for $i = 1, \dots, m$ and $c = 1, \dots, k$, which follows $G(\beta, \gamma)$. Based on Equations (1.1) and (1.2), the PDF of X_{ic} is:

$$g(x_{ic} \mid \beta, \gamma) = \sum_{r=0}^{i-1} i \binom{m}{i} \binom{i-1}{r} (-1)^r \beta e^{\gamma x_{ic}} e^{-\frac{\beta}{\gamma} (e^{\gamma x_{ic}} - 1)(m+r-i+1)}. \quad (2.4)$$

Considering $\mathbf{x} = (x_{1c}, \dots, x_{mc})$, the joint density is:

$$\begin{aligned} g(\mathbf{x} \mid \beta, \gamma) &= \prod_{c=1}^k \prod_{j=1}^m g(x_{jc} \mid \beta, \gamma) \\ &= \prod_{c=1}^k \left\{ \sum_{i_1=0}^0 \sum_{i_2=0}^1 \cdots \sum_{i_m=0}^{m-1} \left[\prod_{j=1}^m j \binom{m}{j} \binom{j-1}{i_j} (-1)^{i_j} \right] \right. \\ &\quad \left. \times \beta^m e^{\gamma \sum_{j=1}^m x_{jc}} e^{-\frac{\beta}{\gamma} \sum_{j=1}^m (m+i_j-j+1)(e^{\gamma x_{jc}} - 1)} \right\}. \end{aligned} \quad (2.5)$$

From Equations (2.5) and the joint prior distribution, the joint posterior distribution for RSS is:

$$\begin{aligned} \pi(\beta, \gamma \mid \mathbf{x}) &\propto \prod_{c=1}^k \left\{ \sum_{i_1=0}^0 \cdots \sum_{i_m=0}^{m-1} \prod_{j=1}^m j \binom{m}{j} \binom{j-1}{i_j} (-1)^{i_j} \right. \\ &\quad \left. \times \beta^{m+a_1-1} e^{-b_1 \beta} \gamma^{a_2-1} e^{-\gamma[b_2 - \sum_{j=1}^m x_{jc}]} e^{-\frac{\beta}{\gamma} \sum_{j=1}^m (e^{\gamma x_{jc}} - 1)(m+i_j-j+1)} \right\}. \end{aligned} \quad (2.6)$$

This equation does not have a known distribution form, so we use MCMC to sample from it. Note that:

$$\begin{aligned} \pi(\beta \mid \mathbf{x}, \gamma) &\propto \prod_{c=1}^k \left\{ \sum_{i_1=0}^0 \cdots \sum_{i_m=0}^{m-1} \prod_{j=1}^m j \binom{m}{j} \binom{j-1}{i_j} (-1)^{i_j} \right. \\ &\quad \left. \times \beta^{m+a_1-1} e^{-\beta[b_1 + \frac{1}{\gamma} \sum_{j=1}^m (e^{\gamma x_{jc}} - 1)(m+i_j-j+1)]} \right\}, \end{aligned} \quad (2.7)$$

$$\begin{aligned} \pi(\gamma \mid \mathbf{x}, \beta) &\propto \prod_{c=1}^k \left\{ \sum_{i_1=0}^0 \cdots \sum_{i_m=0}^{m-1} \prod_{j=1}^m j \binom{m}{j} \binom{j-1}{i_j} (-1)^{i_j} \right. \\ &\quad \left. \times \gamma^{a_2-1} e^{-\gamma[b_2 - \sum_{j=1}^m x_{jc}]} e^{-\frac{\beta}{\gamma} \sum_{j=1}^m (e^{\gamma x_{jc}} - 1)(m+i_j-j+1)} \right\}. \end{aligned} \quad (2.8)$$

3 Estimation of Stress-Strength Reliability

Consider two independent random variables: $X \sim G(\beta_1, \gamma)$ and $Y \sim G(\beta_2, \gamma)$. The stress-strength reliability R is:

$$R = P(X < Y) \quad (3.1)$$

$$\begin{aligned} &= \int_0^\infty F_X(y) f_Y(y) dy \\ &= \int_0^\infty (1 - e^{-\frac{\beta_1}{\gamma}(e^{\gamma y} - 1)}) (\beta_2 e^{\gamma y} e^{-\frac{\beta_2}{\gamma}(e^{\gamma y} - 1)}) dy \\ &= 1 - \int_0^\infty \beta_2 e^{\gamma y} e^{-\frac{1}{\gamma}(e^{\gamma y} - 1)(\beta_1 + \beta_2)} dy \\ &= 1 - \frac{\beta_2}{\beta_1 + \beta_2} = \frac{\beta_1}{\beta_1 + \beta_2}. \end{aligned} \quad (3.2)$$

3.1 R Estimation Based on SRS

Let $\mathbf{X} = (X_1, \dots, X_n)$ and $\mathbf{Y} = (Y_1, \dots, Y_m)$ be two independent SRS from X and Y . The likelihood function for $\boldsymbol{\theta} = (\beta_1, \beta_2, \gamma)$ is:

$$L(\boldsymbol{\theta} \mid \mathbf{x}, \mathbf{y}) \propto \beta_1^n \beta_2^m e^{\gamma(\sum x_i + \sum y_j)} e^{-\frac{1}{\gamma}[\beta_1(\sum e^{\gamma x_i} - n) + \beta_2(\sum e^{\gamma y_j} - m)]}. \quad (3.3)$$

We assume independent priors: $\beta_1 \sim \text{Gamma}(a_1, b_1)$, $\beta_2 \sim \text{Gamma}(a_2, b_2)$, $\gamma \sim \text{Gamma}(a_3, b_3)$.

The joint posterior is:

$$\begin{aligned} \pi(\boldsymbol{\theta} \mid \mathbf{x}, \mathbf{y}) &\propto \beta_1^n \beta_2^m e^{\gamma(\sum x_i + \sum y_j)} e^{-\frac{1}{\gamma}[\beta_1(\sum e^{\gamma x_i} - n) + \beta_2(\sum e^{\gamma y_j} - m)]} \\ &\times \beta_1^{a_1-1} e^{-b_1 \beta_1} \beta_2^{a_2-1} e^{-b_2 \beta_2} \gamma^{a_3-1} e^{-b_3 \gamma}. \end{aligned} \quad (3.4)$$

The full conditional distributions are:

$$(\beta_1 \mid \beta_2, \gamma, \mathbf{x}, \mathbf{y}) \sim \text{Gamma}\left(n + a_1, b_1 + \frac{1}{\gamma} \sum_{i=1}^n (e^{\gamma x_i} - 1)\right), \quad (3.5)$$

$$(\beta_2 \mid \beta_1, \gamma, \mathbf{x}, \mathbf{y}) \sim \text{Gamma}\left(m + a_2, b_2 + \frac{1}{\gamma} \sum_{j=1}^m (e^{\gamma y_j} - 1)\right), \quad (3.6)$$

$$\pi(\gamma \mid \beta_1, \beta_2, \gamma, \mathbf{x}, \mathbf{y}) \propto \gamma^{a_3-1} e^{\gamma(\sum x_i + \sum y_j - b_3)} e^{-\frac{1}{\gamma}[\beta_1 \sum (e^{\gamma x_i} - 1) + \beta_2 \sum (e^{\gamma y_j} - 1)]}. \quad (3.7)$$

3.2 R Estimation Based on RSS

Let $\mathbf{X} = \{X_{ic}, i = 1, \dots, m_x, c = 1, \dots, k_x\}$ and $\mathbf{Y} = \{Y_{jr}, j = 1, \dots, m_y, r = 1, \dots, k_y\}$ be two independent RSS samples from Gompertz(β_1, γ) and Gompertz(β_2, γ), respectively. The joint posterior distribution is:

$$\pi(\boldsymbol{\theta} \mid \mathbf{x}, \mathbf{y}) \propto g(\mathbf{x} \mid \beta_1, \gamma) g(\mathbf{y} \mid \beta_2, \gamma) \pi(\beta_1) \pi(\beta_2) \pi(\gamma).$$

Due to the complexity of the full conditional distributions, we use the Metropolis-Hastings algorithm with proposal $q(\boldsymbol{\theta}) \propto q(\beta_1)q(\beta_2)q(\gamma)$, where $q(\beta_1), q(\beta_2), q(\gamma)$ are independent half-normal densities. The sampling procedure:

Step 1: Initialize $\boldsymbol{\theta}^{(0)} = (\beta_1^{(0)}, \beta_2^{(0)}, \gamma^{(0)})'$, set $t = 1$.

Step 2: Generate $\beta_1^{(t)}$ from $q(\beta_1)$ and form $\boldsymbol{\theta}^{(t)} = (\beta_1^{(t)}, \beta_2^{(t-1)}, \gamma^{(t-1)})'$. Accept with probability

$$\rho = \min \left\{ \frac{\pi(\boldsymbol{\theta}^{(t)} | \mathbf{x}, \mathbf{y})}{\pi(\boldsymbol{\theta}^{(t-1)} | \mathbf{x}, \mathbf{y})} \cdot \frac{q(\boldsymbol{\theta}^{(t-1)} | \boldsymbol{\theta}^{(t)})}{q(\boldsymbol{\theta}^{(t)} | \boldsymbol{\theta}^{(t-1)})}, 1 \right\}.$$

Step 3: Generate $\beta_2^{(t)}$ from $q(\beta_2)$ and set $\boldsymbol{\theta}^{(t)} = (\beta_1^{(t)}, \beta_2^{(t)}, \gamma^{(t-1)})'$, then accept using ρ .

Step 4: Generate $\gamma^{(t)}$ from $q(\gamma)$ and set $\boldsymbol{\theta}^{(t)} = (\beta_1^{(t)}, \beta_2^{(t)}, \gamma^{(t)})'$, then accept using ρ .

Step 5: Compute $R^{(t)} = \beta_1^{(t)} / (\beta_1^{(t)} + \beta_2^{(t)})$.

Step 6: Increment t and repeat Steps 2-5 N times.

The Bayes estimators under SEL loss is:

$$\hat{R}_{\text{SEL}} = \frac{1}{N - M} \sum_{t=M+1}^N R^{(t)}.$$

4 Simulation Study

We conducted Monte Carlo simulations to compare the performance of Bayesian estimators of stress-strength reliability $R = P(X < Y)$ for the Gompertz distribution under SRS and RSS. The simulation parameters were set as follows: number of Monte Carlo runs = 300, MCMC iterations = 6000 with 1000 burn-in, number of cycles $k_x = k_y = 10$, and non-informative priors with hyperparameters 0.0001. Three scenarios for R were considered: $R = 0.2$ ($\beta_1 = 0.25, \beta_2 = 1.0$), $R = 0.5$ ($\beta_1 = 0.5, \beta_2 = 0.5$), and $R = 0.8$ ($\beta_1 = 1.0, \beta_2 = 0.25$), with $\gamma = 0.5$ fixed. Different set sizes $(m_x, m_y) = (2, 2), (3, 3), (4, 3), (4, 4)$ were examined. The mean squared error (MSE), bias, and 95% highest posterior density (HPD) credible interval length were computed for each scenario.

Table 1 presents the MSEs, biases, and HPD lengths of the Bayesian estimates for R under squared error loss (SEL). The results reveal that the performance of RSS relative to SRS depends on the symmetry of the stress and strength distributions. When $R = 0.5$, corresponding to symmetric distributions, RSS substantially outperforms SRS. The relative efficiency reaches up to 3.75 for $R = 0.5$, demonstrating the exceptional performance of RSS under symmetry. For asymmetric scenarios ($R = 0.2$ and $R = 0.8$), SRS shows slightly lower MSE compared to RSS. However, the differences between the two methods are relatively small. More importantly, RSS offers significant practical advantages: it requires only the ranking of units (which is inexpensive and easy) rather

than the precise measurement of all units. This cost-effective feature makes RSS an attractive alternative even when its MSE is slightly higher than SRS. Furthermore, as set sizes increase, the performance gap between SRS and RSS narrows, and the HPD interval lengths remain comparable across both methods.

Table 1: MSEs, biases, and HPD lengths of Bayesian estimates for R under SEL

R	m_x	m_y	MSE(SRS)	MSE(RSS)	Bias(SRS)	Bias(RSS)	HPD(SRS)	HPD(RSS)
0.2	2	2	0.0061	0.0083	0.0326	0.0798	0.0070	0.0075
0.2	3	3	0.0044	0.0073	0.0168	0.0810	0.0056	0.0059
0.2	4	3	0.0029	0.0093	0.0079	0.0926	0.0050	0.0056
0.2	4	4	0.0025	0.0074	0.0092	0.0833	0.0048	0.0051
0.5	2	2	0.0060	0.0024	0.0121	0.0066	0.0089	0.0088
0.5	3	3	0.0050	0.0010	-0.0032	-0.0069	0.0072	0.0072
0.5	4	3	0.0034	0.0010	-0.0063	0.0044	0.0068	0.0067
0.5	4	4	0.0030	0.0008	-0.0035	-0.0073	0.0063	0.0063
0.8	2	2	0.0041	0.0064	-0.0103	-0.0682	0.0066	0.0074
0.8	3	3	0.0026	0.0094	-0.0176	-0.0930	0.0057	0.0060
0.8	4	3	0.0022	0.0077	-0.0176	-0.0842	0.0054	0.0055
0.8	4	4	0.0019	0.0095	-0.0137	-0.0951	0.0049	0.0052

Discussion and Conclusion

This paper showed that Ranked set sampling significantly improves the estimation of $R = P(X < Y)$ under the Gompertz distribution compared to simple random sampling, especially when distributions are symmetric. The proposed Bayesian estimators are both reliable and economical, making RSS a practical choice for engineering applications with costly measurements.

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